

Integrals Within Integrals

A Formalization of the Gagliardo–Nirenberg–Sobolev Inequality

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Based on the 2024 paper by Floris van Doorn and Heather Macbeth
<https://drops.dagstuhl.de/entities/document/10.4230/LIPIcs.ITP.2024.37>

1 Motivation

Analysis is severely underrepresented in mathlib. Reasons behind this are both mathematical (more “techniques”) and sociological (less people doing Lean interested in it).

One peculiar aspect are integrals: Mathlib knows two kinds of integrals, the lower Lebesgue integral as well as the Bochner integral.

On top of this, formally doing integration can be tricky; for example often one has to prove integrability or measurability of functions.

Fubini and Tonelli theorem roughly tell us that, when integrating over two variables, the order of integration does not matter.

This is perfectly provable in Lean, but when implemented “naively” can be hard to work with. Roughly speaking, this is mainly due to $\mathbb{R}^a \times \mathbb{R}^b$ not being (definitionally) equal to $\mathbb{R}^{a+b}$. So to work with iterated integrations without special machinery, one has to (un)curry functions and carry around type equivalences.

2 Marginals

The marginal construction from the paper is meant to be the answer to this problem.

Let $(X_i, \mu_i)_{i \in \iota}$ be measure spaces, let $A \subseteq \iota$ be finite, and let $f : (\prod_i X_i) \rightarrow [0, \infty]$. The *marginal of f over A* integrates exactly the coordinates indexed by A :

$$\int \cdots \int_{i \in A} f d\mu_i(x) := \int_{\prod_{i \in A} X_i} f(X_A(x, y)) d\left(\prod_{i \in A} \mu_i\right)(y), \quad X_A(x, y)_i = \begin{cases} y_i, & i \in A, \\ x_i, & i \notin A. \end{cases}$$

Crucially, every marginal again has type $(\prod_i X_i) \rightarrow [0, \infty]$, independently of A . This might be a bit unintuitive at first (i.e. integrating over all variables gives a constant function and not a value), but drastically simplifies (iterated) integration in practice, by eliminating the need to ever have to worry about things like (un)currying and dealing with type equivalences.

In Lean:

```
def lmarginal (μ : ∀ i, Measure (X i)) (s : Finset ι)
  (f : (∀ i, X i) → ENNReal) (x : ∀ i, X i) : ENNReal :=
  ∫- y : ∀ i : s, X i,
  f (updateFinset x s y) ∂ Measure.pi (fun i : s => μ i)
```

Here, `updateFinset x s y` replaces the coordinates of x belonging to s by the corresponding values of y .

Typically one will not work with the definition directly too much, but instead rely on some basic identities (which are easy to prove by unfolding the definition), for example

$$\int \cdots \int_{i \in \emptyset} f d\mu_i = f, \quad \int \cdots \int_{i \in \{i\}} f d\mu_i(x) = \int_{X_i} f(x[i \mapsto t]) d\mu_i(t) \quad \int \cdots \int_{i \in \iota} f d\mu_i = \int_X f d\sqcap_{i \in \iota} \mu_i$$

3 Tonelli’s theorem in coordinate form

For measurable f and disjoint finite sets $A, B \subseteq \iota$, Tonelli’s theorem becomes

$$\int \cdots \int_{i \in A \cup B} f d\mu_i = \int \cdots \int_{i \in A} \left(\int \cdots \int_{i \in B} f d\mu_i \right) d\mu_i.$$

The proof identifies $(\prod_{i \in A} X_i) \times (\prod_{i \in B} X_i)$ with $\prod_{i \in A \cup B} X_i$ by a measurable equivalence and then applies the usual Tonelli theorem. The resulting Lean interface hides this bookkeeping:

```

theorem lmarginal_union
  (f : (∀ i, X i) → ENNReal)
  (hf : Measurable f) (hst : Disjoint s t) :
  lmarginal μ (s ∪ t) f
    = lmarginal μ s (lmarginal μ t f) := by
  ...

```

4 Applications

4.1 Volume of an n -ball

Marginals work well with induction (for examples with `Finset.induction`) and allow one to integrate one coordinate at a time. This is suitable for concrete volume calculations like n -spheres. Here if we denote

$$B_n = \frac{\pi^{n/2}}{\Gamma(n/2 + 1)},$$

by induction + 1 dimensional integration one obtains

$$\text{vol}\{x \in \mathbb{R}^n : \|x\| \leq R\} = B_n R^n$$

thus proving that B_n is in fact the volume of a (unit) n -sphere. Many other volume calculation can be carried out in a similar fashion with marginals.

4.2 Gagliardo–Nirenberg–Sobolev inequality

The main application of the marginal construction in the paper is the proof of the Gagliardo–Nirenberg–Sobolev inequality:

Let E be an n -dimensional real normed space, let $1 \leq p < n$, and set

$$p^* = \frac{np}{n-p}.$$

Then there exists $C \geq 0$ such that every compactly supported C^1 function $u : E \rightarrow \mathbb{R}$ satisfies

$$\|u\|_{L^{p^*}} \leq C \|Du\|_{L^p}.$$

While the inequality is a rather classic result, in the literature it is for reasons of simplicity typically only proved for $n = 2$ or $n = 3$ (but stated for general n).

So to come up with a formal proof in Lean, Floris and Heather first had to generalise the proofs from the literature. A technical lemma, they dubbed the “grid-lines” estimate, became the main piece of the proof

```

theorem lintegral_mul_prod_lintegral_pow_le
  (hp : (#ι - 1 : ℝ) * p ≤ 1)
  {f : (∀ i, A i) → ENNReal} (hf : Measurable f) :
  ∫- x, f x ^ (1 - (#ι - 1 : ℝ) * p)
    * product i, (∫- xi, f (update x i xi) ∂ μ i) ^ p
    ∂ Measure.pi μ
  ≤ (∫- x, f x ∂ Measure.pi μ) ^ (1 + p) := by
  ...

```

The Gagliardo–Nirenberg–Sobolev inequality is important for the theory of Sobolev spaces, for example via the Sobolev embedding theorem, and PDEs in general.

It can be used to make statements about (weak) solutions of PDEs, which we can not solve explicitly.

While there is no PDE theory in `mathlib` yet, the main prerequisites are at least in the PR queue, for example Sobolev space in #32305.

Furthermore, there have been vibe-coded PDE projects in Lean, further showcasing that a proper formalisation of differential equations is feasible.

5 Future work

While marginals are very nice to work with, they are barely used in `mathlib` at all.

While it is plausible that the marginal construction might be more relevant in concrete calculation, the main reasons is for the low popularity might simply just be that nobody cared to golf proofs and statements.

Additionally, the marginal constructions as of now only consists for the lower Lebesgue integral, but not Bochner integrals.

Surprisingly “translating” the work is not a trivial task (as it requires some additional definitions to be useful). Currently being worked on by Joris Roos and Floris van Doorn.